



UNIVERSITY OF WESTERN ATTICA
DEPARTMENT OF COMPUTER AND
INFORMATION TECHNOLOGY AND
COMPUTER ENGINEERING
PHYSICS

PROJECT 1
EXERCISE IN ERRORS AND
GRAPHICS

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SECTION 1

SEMESTER: 1 °

ANSWERS

1) n number (number of p.s.)

976,45 (5)

10 (2)

86400 (5)

6400 (4)

5280 (4)

0,00094 (2)

$4,00 \times 10^2$ (3)

0,640 (3)

2) n s.p. -> 2 s.p.

8,314 -> 8,3

$4,00123 \times 10^3$ -> 0,0040

1,03 -> 1,0

9,9999 -> 10 2,045 -> 2,0

55,5555 -> 56

99,5 -> 10 X 10

1,024 -> 1,0

3) Total counts are 10 (n<25)

$$\bar{x} \pm \sigma(\bar{x}) = (2,34 \pm 0,06) \text{ mm}$$

$$\bar{u} \pm \sigma(\bar{u}) = (9,0 \pm 0,1) \times 10 \text{ m/s}$$

4) Tables 1 and 2 are completed in the file "EXERCISE TABLES

4.excel"

TABLE 1

$$\bar{x} = Sx_i / n \Rightarrow \bar{x} = 283.3/5 \Rightarrow$$

$$\bar{x} = 56,66 \text{ cm}$$

$$\sigma(x^-) = \sqrt{\{\Sigma(\Delta x_i)^2/n(n-1)\}} \Rightarrow \sigma(x^-) = \sqrt{(0.372/20)} \Rightarrow$$

$$\sigma(x^-) = 0.136381817 \text{ cm } \sigma_x \% = (\sigma(x^-)/x^-) \times 100 \Rightarrow \sigma_x \% =$$

$$(0.136381817/56.66) \times 100 \Rightarrow$$

$$\sigma_x \% = 0,2407021126$$

Total counts are 5 ($n < 25$)

$$x^- \pm \sigma(x^-) = (56.7 \pm 0.1) \text{ cm}$$

$$x^- \pm \sigma_x \% = (56.7 \text{ cm} \pm 0.2\%)$$

TABLE 2

$$y^- = \Sigma y_i / n \Rightarrow y^- = 86.5/5 \Rightarrow$$

$$y^- = 17,3 \text{ cm}$$

$$\sigma(y^-) = \sqrt{\{\Sigma(\Delta y_i)^2/n(n-1)\}} \Rightarrow \sigma(y^-) = \sqrt{(0.10/20)} \Rightarrow$$

$$\sigma(y^-) = 0,0707106781 \text{ cm}$$

$$\sigma_y \% = (\sigma(y^-)/y^-) \times 100 \Rightarrow \sigma_y \% = (0,0707106781/17,3) \times 100 \Rightarrow$$

$$\sigma_y \% = 4,0873224335$$

Total measurements are 5 ($n < 25$)

$$y^- \pm \sigma(y^-) = (17,3 \pm 0,07) \text{ cm}$$

$$y^- \pm \sigma_y \% = (17,3 \text{ cm} \pm 4\%)$$

Area A = area of rectangular rectangle

$$\bar{A} = x^- \times y^- \Rightarrow \bar{A} = 56,66 \times 17,3 \Rightarrow$$

$$\bar{A} = 980,218 \text{ cm}^2$$

$$\sigma(\bar{A}) = \sigma(x^-) \times \sigma(y^-) \Rightarrow \sigma(\bar{A}) = 0,136381817 \times 0,0707106781 \Rightarrow$$

$$\sigma(\bar{A}) = 0,0096436508 \text{ cm}^2$$

$$\sigma_A \% = \sigma_x \% \times \sigma_y \% \Rightarrow \sigma_A \% = 0,2407021126 \times 4,0873224335 \Rightarrow$$

$$\sigma \%_A = 0,9838271446$$

The total number of measurements is 10 ($n < 25$)

$$\bar{A} \pm \sigma(\bar{A}) = (980.22 \pm 0.01) \text{ cm}^2 \bar{A}$$

$$\pm \sigma_A \% = (980.2 \text{ cm}^2 \pm 1.0\%)$$

5) Total measurements are 10

$$T_i = 2\pi v(x_i + a)/g$$

Least squares method

$$S(x_i \times T_i) = (90,7 \times 2,05) + (80,4 \times 2,01) + (70,6 \times 1,86) + (60,1 \times 1,75) + (49,7 \times 1,62) + (38,9 \times 1,47) + (28,9 \times 1,34) + (17,9 \times 1,24) + (7,8 \times 0,96) + (5,6 \times 0,92) \Rightarrow$$

$$\underline{\Sigma(x_i \times T_i) = 795,289 \text{ cm} \times \text{s}}$$

$$\Sigma(x_i) = 90,7 + 80,4 + 70,6 + 60,1 + 49,7 + 38,9 + 28,9 + 17,9 + 7,8 + 5,6 \Rightarrow$$

$$\underline{\Sigma(x_i) = 450,6 \text{ cm}}$$

$$\Sigma(T_i) = 2,05 + 2,01 + 1,86 + 1,75 + 1,62 + 1,47 + 1,34 + 1,24 + 0,96 + 0,92 \Rightarrow$$

$$\underline{\Sigma(T_i) = 15,22 \text{ s}}$$

$$\Sigma(x_i^2) = (90,7)^2 + (80,4)^2 + (70,6)^2 + (60,1)^2 + (49,7)^2 + (38,9)^2 + (28,9)^2 + (17,9)^2 + (7,8)^2 + (5,6)^2 \Rightarrow$$

$$\underline{\Sigma(x_i^2) = 28518,14 \text{ cm}^2}$$

$$(\Sigma x_i)^2 = \Sigma(x_i) \times \Sigma(x_i) \Rightarrow (\Sigma x_i)^2 = 450,6 \times 450,6 \Rightarrow$$

$$\underline{(\Sigma x_i)^2 = 203040,36 \text{ cm}^2}$$

$$a = n \times \Sigma(x_i \times T_i) - \Sigma(x_i) \times \Sigma(T_i) / n \times \Sigma(x_i^2) - (\Sigma x_i)^2 \Rightarrow$$

$$\alpha = 10 \times 795,289 - 450,6 \times 15,22 / 10 \times 28518,14 - 203040,36 \Rightarrow$$

$$\alpha = \mathbf{0,0133277835 \text{ cm}}$$

$$g = \Sigma(T_i) \times \Sigma(x_i^2) - \Sigma(x_i) \times \Sigma(x_i \times T_i) / n \times \Sigma(x_i^2) - (\Sigma x_i)^2 \Rightarrow$$

$$g = 15,22 \times 28518,14 - 450,6 \times 795,289 / 10 \times 28518,14 - 203040,36 \Rightarrow$$

$$g = \mathbf{0,9214500742 \text{ cm}^2/\text{s}}$$

6) Total measurements are 10

$$T_i = 2\pi v/m_i + m_{e\lambda} / D$$

Least squares method

$$\Sigma(T_i) = 0,396 + 0,536 + 0,647 + 0,741 + 0,824 + 0,900 + 0,970 + 1,04 + 1,10 + 1,15 \Rightarrow$$

$$\underline{\Sigma(T_i) = 8,304 \text{ s}}$$

$$\Sigma(m_i) = 0,05 + 0,10 + 0,15 + 0,20 + 0,25 + 0,30 + 0,35 + 0,40 + 0,45 + 0,50 \Rightarrow$$

$$\underline{\Sigma(m_i) = 2,75 \text{ kg}}$$

$$\Sigma(m_i \times T_i) = (0,05 \times 0,396) + (0,10 \times 0,536) + (0,15 \times 0,647) + (0,20 \times 0,741) + (0,25 \times 0,824) + (0,30 \times 0,900) + (0,35 \times 0,970) + (0,40 \times 1,04) + (0,45 \times 1,10) + (0,50 \times 1,15) \Rightarrow$$

$$\underline{\Sigma(m_i \times T_i) = 2,60035 \text{ kg} \times \text{s}}$$

$$\Sigma(m_i^2) = (0,05)^2 + (0,10)^2 + (0,15)^2 + (0,20)^2 + (0,25)^2 + (0,30)^2 + (0,35)^2 + (0,40)^2 + (0,45)^2 + (0,50)^2 \Rightarrow$$

$$\underline{\Sigma(m_i^2) = 0,9625 \text{ kg}^2}$$

$$(\Sigma m_i)^2 = \Sigma(m_i) \times \Sigma(m_i) \Rightarrow$$

$$\underline{(\Sigma m_i)^2 = 7,5625 \text{ kg}^2}$$

$$m_{\varepsilon\lambda} / 3 = n \times \Sigma(m_i \times T_i) - \Sigma(m_i) \times \Sigma(T_i) / n \times \Sigma(m_i^2) - (\Sigma m_i)^2 \Rightarrow$$

$$m_{\varepsilon\lambda} / 3 = 10 \times 2,60035 - 2,75 \times 8,304 / 10 \times 0,9625 - 7,5625 \Rightarrow$$

$$m_{\varepsilon\lambda} = 3 \times 1,5357575758 \Rightarrow$$

$$\mathbf{m_{\varepsilon\lambda} = 4,6072727273 \text{ kg}}$$

$$D = \Sigma(T_i) \times \Sigma(m_i^2) - \Sigma(m_i) \times \Sigma(m_i \times T_i) / n \times \Sigma(m_i^2) - (\Sigma m_i)^2 \Rightarrow$$

$$D = 8,304 \times 0,9625 - 2,75 \times 2,60035 / 10 \times 0,9625 - 7,5625 \Rightarrow$$

$$\mathbf{D = 0,4080666667 \text{ N/m}}$$